Scalability & Future Plan

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Team ID	Individual Project — Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Data is a static 300,000-row extract; it does not refresh automatically	Insights go out of date as new market data appears	High
2	Analysis is descriptive only; there is no forecasting or prediction (for example resale value or battery degradation)	Stakeholders cannot plan for future sales or prices	Medium
3	Dashboard is public on Tableau Public and the web app depends on its embed; there is no login or access control	Not suitable for confidential data; the app is affected if the embed is unavailable	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask app on Vercel embedding a public Tableau dashboard; suitable for a small number of viewers	Move to Tableau Cloud or Server for managed capacity; use Vercel caching/CDN for the web app
2	Data Storage	CSV files combined into a Tableau extract (300,000 rows × 37 columns)	Store data in a cloud data warehouse (for example Snowflake or BigQuery) with scheduled refresh
3	Performance	Filters and cross-filtering respond in about 1–2 seconds using an extract	Pre-aggregate large tables, limit marks per view and monitor with Tableau performance recording
4	Security	Public Tableau Public link, no authentication	Private dashboards, user logins and role-based views; no confidential data on public servers

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Connect to a live data source or warehouse with automatic refresh	Within 1–2 months	Always up-to-date insights without manual extract updates
Phase 3	Machine learning models for resale value, battery degradation and sales forecasting	Within 3–4 months	Move from descriptive to predictive decision support
Phase 4	User login, role-based views and price/efficiency analysis	Within 5–6 months	Secure multi-user access and deeper product positioning

			insights